

Electronics Design with VHDL.

Objects, data types and operators (V.0)

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1: Data Type Conversion

Given the signal declaration `signal x : std_logic_vector(3 down to 0) := "1010";`
Indicate the statement to correctly convert x into an integer

2: Logical Operators

Determine which of the following sentences is a valid logical operation in VHDL

- A. `signal a : std_logic; a := a AND '1';`
- B. `signal b : std_logic; b <= b OR '0';`
- C. `signal c : std_logic_vector(3 downto 0); c <= c XOR "1010";`
- D. `signal d : std_logic; d := d NAND '1';`

3: Aggregate Allocation

Determine a valid way to assign all bits of a 4-bit vector `std_logic_vector` signal to '0'

4: Relational Operators

Given two signals a and b of type `std_logic_vector(3 to 0)`, determine an expression that checks whether a is less than b?

5: Matrix Statement

Declare a `std_logic_vector` array1 array with 8 elements, each 4 bits wide

6: Basic Data Types

What will happen when you compile the following VHDL code?

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity BasicDataTypesTest is
end BasicDataTypesTest;

architecture Behavioral of BasicDataTypesTest is
    signal a : bit := '1';
    signal b : std_logic := '0';
begin
    process
    begin
        a <= b;
        wait;
    end process;
end Behavioral;
```

7: Arrays and Vectors

Determine the result of the following VHDL code after simulation

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity ArrayTest is
end ArrayTest;

architecture Behavioral of ArrayTest is
    signal arr : std_logic_vector(3 downto 0) := "1010";

begin
    process
        variable temp : std_logic_vector(3 downto 0);
    begin
        temp := arr(1 downto 0) & arr(3 downto 2);
        wait;
    end process;
end Behavioral;
```

8: Record Types

Determine the incorrect assignments following the next VHDL code

```
type person is record
    name : std_logic_vector(7 downto 0);
    age : integer range 0 to 127;
end record;
signal john : person := (name => "John", age => 25);
```

- A. john.name <= "01001010";
- B. john.age <= 30;
- C. john.age <= -1;
- D. john.name <= (others => '0');

9: Signal and Variable Behaviour

Determine the value of output_signal after the process executes of the following VHDL code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity SignalVarTest is
    port (
        clk : in std_logic;
        output_signal : out std_logic
    );
end SignalVarTest;
```

```

architecture Behavioral of SignalVarTest is
    signal signal_var : std_logic := '0';
begin
    process(clk)
        variable temp_var : std_logic := '0';
    begin
        if rising_edge(clk) then
            temp_var := not temp_var;
            signal_var <= temp_var;
        end if;
        output_signal <= signal_var;
    end process;
end Behavioral;

```

- A. output_signal toggles between '0' and '1' on every rising edge of the clock.
- B. output_signal always remains '0'.
- C. output_signal toggles between '1' and '0' but with one clock cycle delay.
- D. output_signal will be undefined.

10: Data Type Compatibility

Determine the result of the following VHDL code after simulation

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity TypeTest is
end TypeTest;

architecture Behavioral of TypeTest is
    signal std_logic_var : std_logic := '1';
    signal bit_vector_var : bit_vector(3 downto 0) := "1010";
begin
    process
    begin
        std_logic_var <= bit_vector_var(0);
        wait;
    end process;
end Behavioral;

```

- A. The code will compile and simulate successfully, assigning '0' to std_logic_var.
- B. The code will compile, but a runtime error will occur due to type mismatch.
- C. The code will not compile due to a type mismatch error.
- D. The code will compile and assign '1' to std_logic_var.

11: Arithmetic Operations with Different Data Types

Determine the result of the following VHDL code, and the result_signal after the process executes

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

```

```
entity ArithmeticTest is
end ArithmeticTest;
```

architecture Behavioral of ArithmeticTest is

```
    signal a : std_logic_vector(3 downto 0) := "0110"; -- 6 in decimal
    signal b : std_logic_vector(3 downto 0) := "0011"; -- 3 in decimal
    signal result_signal : std_logic_vector(4 downto 0);
begin
    process
    begin
        result_signal <= ('0' & a) + ('0' & b);
        wait;
    end process;
end Behavioral;
```

- A. result_signal will hold "10001".
- B. result_signal will hold "00001".
- C. result_signal will hold "01001".
- D. The code will not compile due to an operator error.

12: Logical Operators on std_logic_vector

Determine the result of the following VHDL code, and the value of result_vector

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
```

```
entity LogicalOperationTest is
end LogicalOperationTest;
```

architecture Behavioral of LogicalOperationTest is

```
    signal vec1 : std_logic_vector(3 downto 0) := "1101";
    signal vec2 : std_logic_vector(3 downto 0) := "1011";
    signal result_vector : std_logic_vector(3 downto 0);
begin
    process
    begin
        result_vector <= (vec1 and vec2) or (vec1 nand vec2);
        wait;
    end process;
end Behavioral;
```

13: Aggregate Assignments

Determine the result of the following VHDL code (assume output_vector is defined as a 4-bit vector)

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
```

```
entity AggregateAssignmentTest is
end AggregateAssignmentTest;
```

```

architecture Behavioral of AggregateAssignmentTest is
    signal input_vector : std_logic_vector(3 downto 0) := "1101";
    signal output_vector : std_logic_vector(3 downto 0);
begin
    process
    begin
        output_vector <= (others => input_vector(0));
    end process;
end Behavioral;

```

14: What is the difference between std_logic_vector type and signed and unsigned types?

15: Determine library and package that should we include in a VHDL design if we want to use signed and unsigned types

16: Determine the difference between whole types and signed and unsigned types

17: What is the difference between the whole type and the positive type?

18: Write VHDL code that converts 8 bit signed signal to a std_logic_vector, Perform the conversion of the resultant std_logic_vector to an unsigned type.

19: Write a VHDL code that converts an integer type to an 8 bit std_logic_vector. The integer should be treated as an unsigned number.

20: Create a VHDL entity and architecture that defines a signal sig1 of type std_logic. Assign a constant value '1' to this signal

21: Create a VHDL entity that defines an std_logic_vector of length 4. Use logical operators to perform a bitwise AND operation between two vectors.

22: Create a VHDL entity that defines an array of type std_logic_vector. Write a process to reverse the bits of the vector using a loop.

23: Create a VHDL entity that performs an arithmetic addition of two 4-bit unsigned numbers and stores the result in a 5-bit vector to handle overflow.

24: Create a VHDL entity that compares two 4-bit vectors using relational operators (<, >, =). Set a signal is_greater to '1' if the first vector is greater than the second.

25: Attributes

1. signal data : std_logic_vector(7 downto 0), what does the data'range attribute return?
2. if (clk'event and clk = '1') then
3. signal vector_signal : std_logic_vector(15 downto 0); What does the vector_signal'length attribute return?
4. signal b : std_logic_vector(3 to 10); What value will the b'high attribute return?
5. What do you get by using the 'delayed(5 ns) attribute on a signal?
6. Define an attribute called my_attribute for a signal and assign it an integer value of 10.